

Transfer Rates -- Guided Notes

Unit tp-1.6 | CompTIA Network+ N10-009 Obj. FC0-U71 1.3 | N10-008 FC0-U71 1.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Network and connection speeds are measured in _____ per second (lowercase b), while file sizes are measured in _____ (uppercase B). To convert from Mbps to MB/s, you divide by _____. A 100 Mbps internet connection transfers data at _____ MB/s.

Question 2

USB 2.0 has a maximum transfer speed of _____ Mbps. USB 3.0 has a maximum of _____ Gbps. USB 3.0 is approximately _____ times faster than USB 2.0. USB 3.0 ports are typically identified by their _____ color inside the port.

Question 3

Calculate how long it would take to transfer a 1 GB file over a USB 2.0 connection at a sustained 480 Mbps:

Convert speed to MB/s: $480 \text{ Mbps} / 8 =$ _____ MB/s

1 GB = _____ MB

Time = $1,024 /$ _____ = approximately _____ seconds

Question 4

_____ refers to the maximum theoretical data transfer capacity of a connection, usually measured in Mbps or Gbps. _____ is the actual measured data transfer rate achieved in real conditions, which is always _____ (greater/less) than the theoretical maximum due to overhead and interference.

Question 5

_____ is the delay between when a signal is sent and when it is received, measured in milliseconds (ms). It does not affect _____ (how much data per second) but affects how responsive a connection feels. High latency causes noticeable delays in _____ gaming, video calls, and remote desktop sessions.

Question 6

An ISP advertises an internet plan as 500 Mbps download. A user downloading a file notices the download progress bar shows around 60 MB/s. Is this accurate or is there a problem? Show the conversion:

500 Mbps / 8 = _____ MB/s

60 MB/s is _____ (above/below/at) the theoretical maximum.

Question 7

Mbps stands for _____ per second. MBps stands for _____ per second. The difference is _____ times larger. Internet service providers advertise in _____ (Mbps/MBps) while download managers typically display progress in _____ (Mbps/MBps).

Question 8 -- Real World Application

REAL WORLD: A student is downloading a 4 GB game on a home internet connection advertised at 200 Mbps. Their download client shows the download speed fluctuating between 15 MB/s and 20 MB/s, averaging about 18 MB/s. They expect much faster speeds based on the 200 Mbps advertisement. Explain why actual speed is lower and calculate the expected download time at 18 MB/s.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-1-6-transfer-rates/tp-1-6-transfer-rates-answer-key.pdf
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